Advanced Machine Learning

Bogdan Alexe,

bogdan.alexe@fmi.unibuc.ro

University of Bucharest, 2nd semester, 2025-2026

Practical Machine Learning (sem 1) …

Lectures

Lecture 1
- Introduction to Machine Learning
- Basic Concepts
- Learning Paradigms

Lecture 2
- Basic Concepts
- Naive Bayes
- Performance Metrics

Lecture 3
- Nearest Neighbors
- Local Learning
- Curse of Dimensionality

Lecture 4
- Decision Trees
- Random Forests

Lecture 5
- Kernel Methods
- Ridge Regression

Lecture 6
- Support Vector Machines
- Logistic Regression

Lecture 7
- Loss Functions and Optimization
- Gradient Descent
- Code

Lecture 8
- Neural Networks
- Introduction to Deep Learning
- Dropout
- Code

Lecture 9
- Convolutional Neural Networks
- Convolutional Layer
- Pooling Layer

Lecture 10
- Bag-of-Words
- Term Frequency - Inverse Document Frequency
- Bag-of-Visual-Words
- Histogram of Oriented Gradients

Lecture 11
- K-means
- Clustering Goodness
- Soft k-means
- Kernel k-means

Lecture 12
- DBSCAN
- Clustering by unmasking
- Hierarchical Clustering

Practical Machine Learning (sem 1) …

Labs

Installing Anaconda – Windows

Installing Anaconda – Linux

Lab 1

- Introduction to Python
- Introduction to Numpy
- Introduction to Matplotlib
- Solution

Lab 2

- K Nearest Neighbors
- Naive Bayes
- Solution

Lab 3

- Kernel Ridge Regression
- Support Vector Machines
- Solution

Lab 4

- Neural Networks
- Convolutional Neural Networks
- Solution

Lab 5

- K-means
- DBSCAN
- Solution

Lab 6

- K-means
- DBSCAN
- Hierarchical Clustering

Projects

Supervised Task on Kaggle

Link to challenge

This competition has finished.

Unsupervised Task on Individual

Link to chosen datasets and methods

Students are required to propose a unique dataset
the lab assitants).

- Lectures and lab classes oriented towards written ML programs in Python

… vs. Theoretical Machine Learning (sem 2)

- Lectures and seminars oriented towards understanding the theory behind the ML algorithms

Example of a slide from a lecture

Theorem 5.1. (No-Free-Lunch) *Let A be any learning algorithm for the task of binary classification with respect to the $0-1$ loss over a domain $\mathcal{X}$. Let m be any number smaller than $|\mathcal{X}|/2$, representing a training set size. Then, there exists a distribution $\mathcal{D}$ over $\mathcal{X} \times \{0,1\}$ such that:*

1. *There exists a function $f : \mathcal{X} \to \{0,1\}$ with $L_{\mathcal{D}}(f) = 0$.*
2. *With probability of at least $1/7$ over the choice of $S \sim \mathcal{D}^m$ we have that $L_{\mathcal{D}}(A(S)) \geq 1/8$.*

… vs. Theoretical Machine Learning (sem 2)

- Lectures and seminars oriented towards understanding the theory behind the ML algorithms

Example of an exercise from seminar

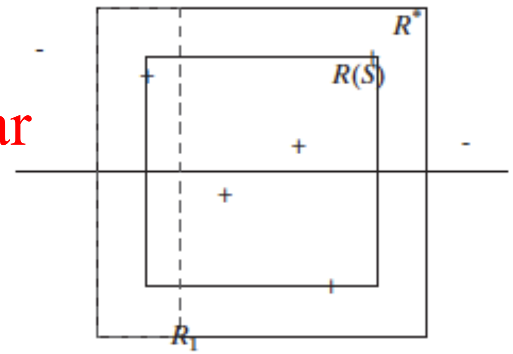


Figure 2.2. Axis aligned rectangles.

Formally, given real numbers $a_1 \le b_1, a_2 \le b_2$, define the classifier $h_{(a_1,b_1,a_2,b_2)}$ by

$$h_{(a_1,b_1,a_2,b_2)}(x_1,x_2) = \begin{cases} 1 & \text{if } a_1 \le x_1 \le b_1 \text{ and } a_2 \le x_2 \le b_2 \\ 0 & \text{otherwise} \end{cases}. \tag{2.10}$$

The class of all axis aligned rectangles in the plane is defined as

$$\mathcal{H}^2_{\mathrm{rec}} = \{h_{(a_1,b_1,a_2,b_2)} : a_1 \le b_1, \text{ and } a_2 \le b_2\}.$$

Note that this is an infinite size hypothesis class. Throughout this exercise we rely on the realizability assumption.

Administrative

You

407 (AI) + 411 (DS) + 412 (NLP)

Optional

+

Nume și prenume student	GRUPA
CIUTAC DARIUS FILIP COSTIN	406
MICLEA ALEXANDRU	406
MUNTEANU MIRIAM	406
Neagoe Mihai Alexandru	406
ORZATA ANDREI	406
POPESCU SEBASTIAN	406
Repciuc Valentin-Adrian	406
BUCA MIHNEA-VICENTIU	408
CHIVEREANU ROBERT-IOAN-ALEXANDF	408
PETRE-SOLDAN ADELA	408
SUMURDUC TEODORA	408
Fabian Andrei	410
NEGRESCU THEODOR	410
ROSOGA RADU-ROLAND	410
SUMURDUC ALEXANDRU	410
Linta Alexandru-Cristian	411

Schedule

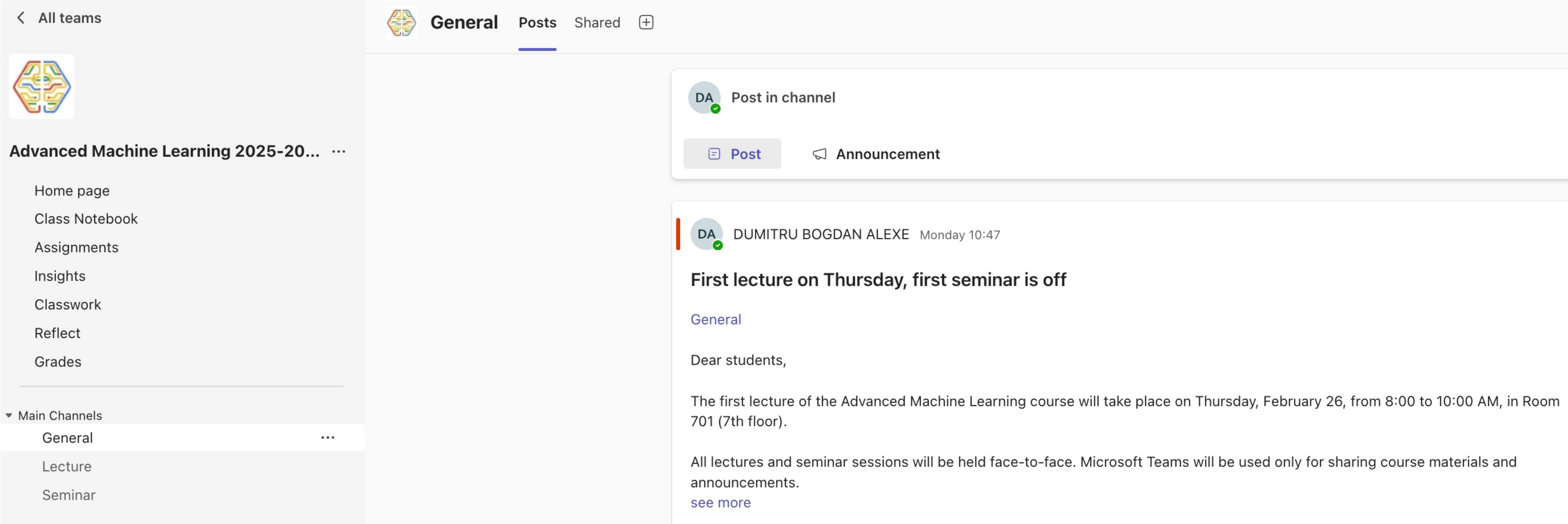
- Lecture : Thursday 8-10 am (weekly), room 701

- Seminar:
  - Thursday 10-12 – room 209
  - Monday 18-20 - room 209
  - in total there will be 6 seminars (one seminar every two weeks)
  - all seminars before the second lecture are off, this Monday + today + next Monday
  - first seminar after the second lecture, next Thursday

F2F vs online

- All lectures and seminar classes are face to face

- I will put the materials for lecture and seminar classes in TEAMS.



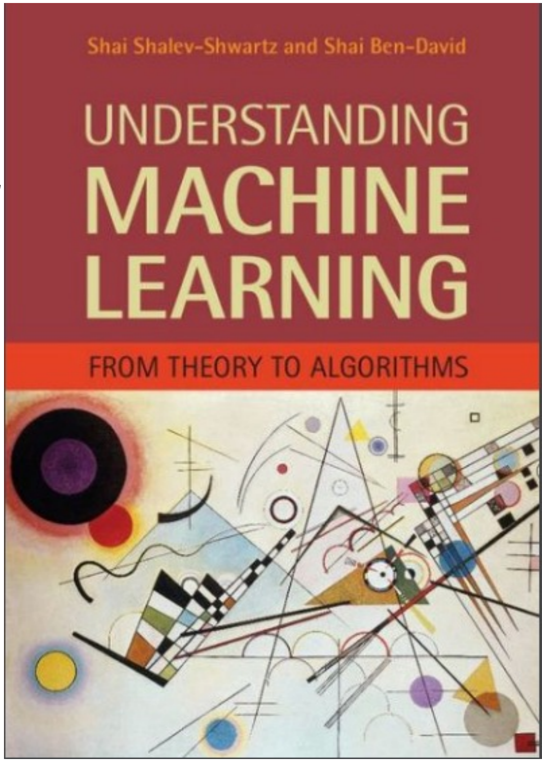
Course Materials

- TEAMS

- Course Book
  - ***Understanding ML from theory to algorithms***

    Shai Shalev-Shwartz, Shai Ben-David,

    Cambridge University Press, 2014
  - **available online**



https://www.cs.huji.ac.il/~shais/UnderstandingMachineLearning/understanding-machine-learning-theory-algorithms.pdf

Course Materials

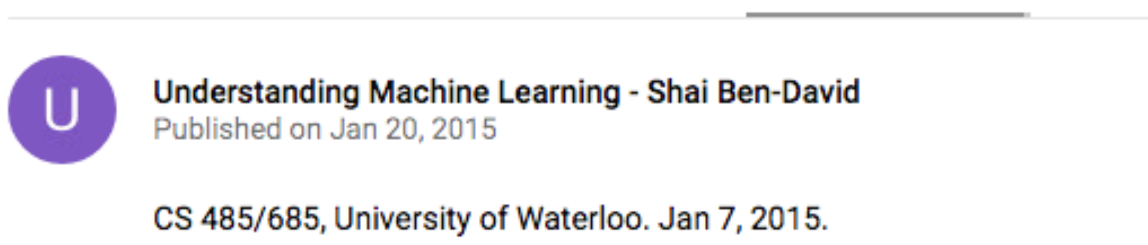
- TEAMS

- Course Book
  - ***Understanding ML from theory to algorithms***

  Shai Shalev-Shwartz, Shai Ben-David,

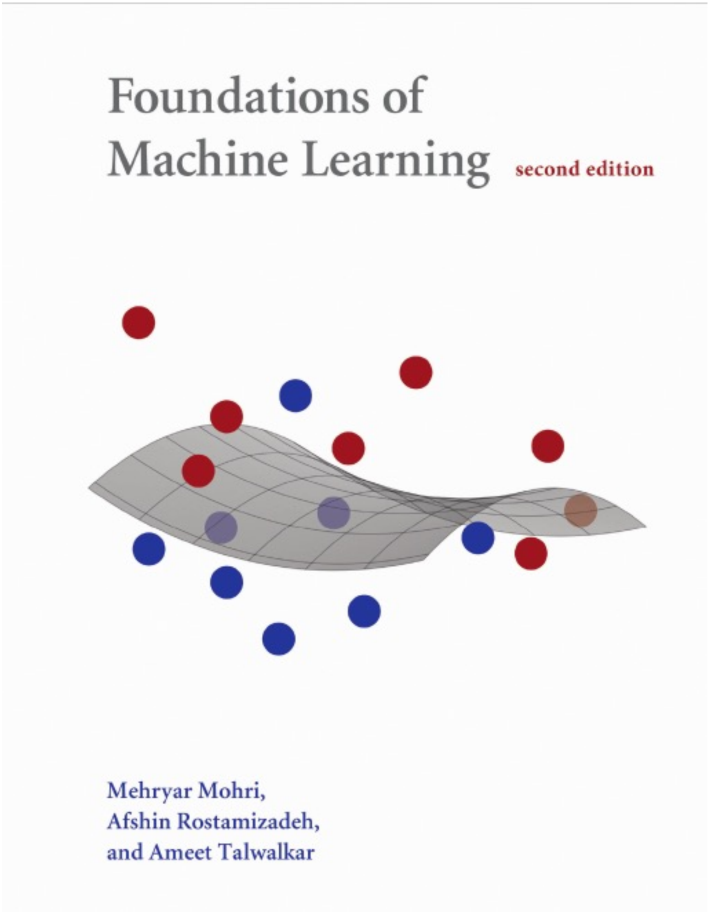
- Online Lectures
  - Shai Ben-David

  Youtube videos

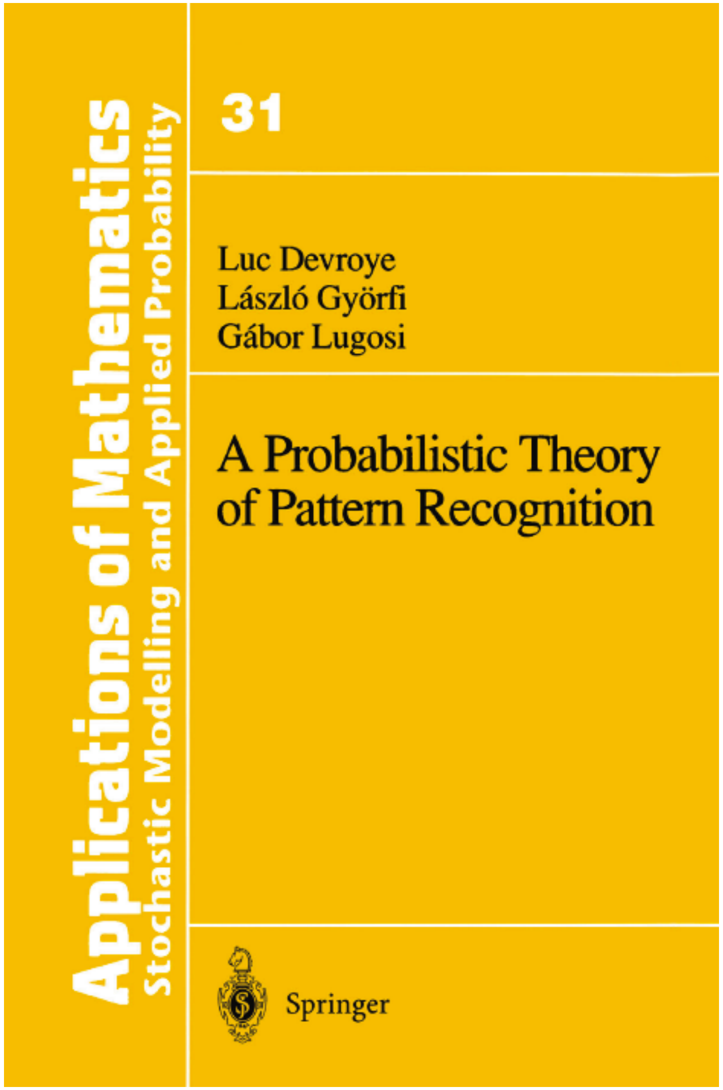


  - Shai Shalev-Shwartz: http://www.cs.huji.ac.il/~shais/IML2014.html

Other Books



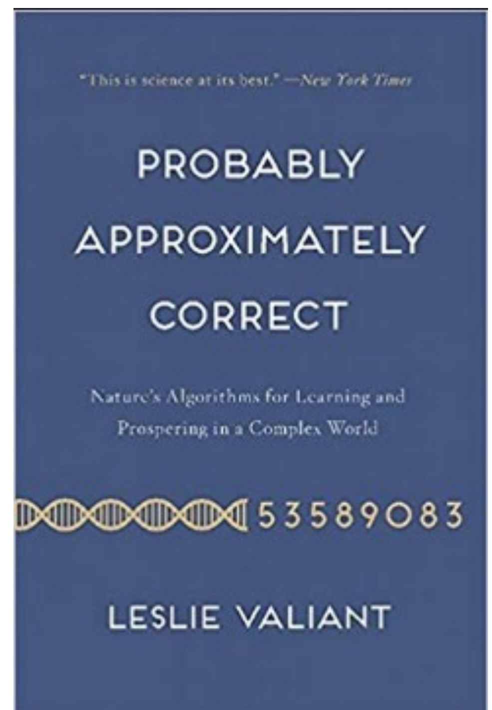
Free online:
https://cs.nyu.edu/~mohri/mlbook/



Other Books

Leslie Valiant, Turing award 2010

For transformative contributions to the theory of computation, including the theory of probably approximately correct (PAC) learning, the complexity of enumeration and of algebraic computation, and the theory of parallel and distributed computing.



Other Books

Leslie Valiant, Turing award 2010

> *For transformative contributions to the theory of computation, including the theory of probably approximately correct (PAC) learning, the complexity of enumeration and of algebraic computation, and the theory of parallel and distributed computing.*

PAC learning (Probably Approximately Correct learning) is a theoretical framework introduced by Leslie Valiant in 1984 that formalizes what it means for a learning algorithm to succeed in a ***distribution free setting:*** the goal is to design an algorithm that, given a set of training examples drawn from an unknown distribution, will produce a hypothesis that is "*probably approximately correct*"—that is, its error is below a pre-specified threshold (ε) – with high probability (at least 1 – δ).

Exam – evaluation in June

Grade = $\min(10, 0.7 \times E1 + \max(A, 0.3 \times E1) + \text{bonus})$

- A = assignment (written) = 3 points
- E1 = written exam face to face, open books = 10 points
- bonus = discuss later
- no constraints, with 4.99 you fail, with 5 you pass

Exam – evaluation in July (retake = restanță)

Grade = min(10, 0.7 × E2 + max(A, 0.3 × E2) + bonus)

- E2 = written exam face to face, open books = 10 points
- bonus = discuss later
- no constraints, with 4.99 you fail, with 5 you pass

Exam – evaluation in September (reexaminare)

Grade = min(10, 0.7 × E3 + max(A, 0.3 × E3) + bonus)

- E3 = written exam face to face, open books = 10 points
- bonus = discuss later
- no constraints, with 4.99 you fail, with 5 you pass

Exam – evaluation in September (mărire)

Grade = min(10, E3 + bonus)

- E3 = written exam face to face, open books = 10 points
- bonus = discuss later

Bonus

- maximum 1 point that you can get as a bonus added to your grade

- only available in this academic year

- Bonus = lecture bonus + seminar bonus

- Lecture bonus:
  - at the end of each lecture you will get a few questions from that lecture. Answer correct and fast to be higher in ranking.
  - we will use Kahoot!
  - based on the Kahoot! ranking at each lecture we will award points
    - students ranked 1-5 get 0.15p
    - students ranked 6-10 get 0.10p
    - students ranked 11-15 get 0.05p

- Seminar bonus:
  - you get 0.1 points per each seminar that you attend (maximum 0.1 points every two weeks), maximum 0.6 points here – easy to achieve ☺

Kahoot - test

- use your phone/laptop to open www.kahoot.com and select PLAY
- use the code given by me
- choose your username, ideally based on your real name
- **read the question on slide, answer on your phone/laptop selecting the right answer**
- quick test!

About assignment

- 2-3 exercises, similar with the ones we solve during the seminar class
- handout in week 7/8, you will have about 2/3 weeks to submit your solution
- late submission policy: maximum 3 days allowed, -10% (= 0.3 points) for each day
- submit your solutions as a pdf written with a scientific text software (Word, Latex, LyX) - this is mandatory, your solution will not be accepted if you submit a handwritten solution
- do not share/copy the solution with/from your colleagues: you + your colleague/s will get 0 points

This is acceptable assignment submission

Solution: The choice between the two approaches is highly dependent on the available data. Since the problem addressed by this algorithm is very sensible, any chosen algorithm may target a very high precision for its predictions.

For analyzing the strengths and weaknesses of each approach we refer to the fact that for each hypothesis h_S we can decompose its error into the sum of an *approximation error* and an *estimation error* [Shai & Shai '14],

$$L_{\mathcal{D}}(h_S) = \epsilon_{\text{app}} + \epsilon_{\text{est}}, \text{ where: } \epsilon_{\text{app}} = \min_{h \in \mathcal{H}} L_{\mathcal{D}}(h) \text{ and } \epsilon_{\text{est}} = L_{\mathcal{D}}(h_S) - \epsilon_{\text{app}}.$$

The advantage of an algorithm that picks axis aligned rectangle in the two dimensional space spanned by the features BP and BMI is that it has a much smaller search space for the hypothesis that reaches the minimum approximation error, ϵ_{app}. We know that the time needed for computing an ERM hypothesis in the class of axis aligned rectangles is quadratic w.r.t. the dimension of search space, therefore this algorithm would run faster than its proposed counterpart. Nevertheless, because the hypothesis space is smaller there are chances that the computed hypothesis will not be exhaustive enough for the general problem. Thus, the approximation error of this algorithm will be higher. Because the estimation error ϵ_{est} will be smaller, this approach can be harder prone to overfitting.

By comparison an algorithm picking axis aligned rectangle in 5 dimensions is more likely to have a smaller approximation error, since there are much more expressive hypothesis available in this class. Still, this algorithm will run slower then the one considering only 2 features and will be also more exposed to overfitting.

This is NOT acceptable assignment submission

a) Din punct de vedere al Bias complexity tradeoff avem următoarele pentru pro și contra referitor la cele 2 modele avem următoarele tipuri de erori.

Eroarea de estimare creste odată cu sample size.

Având mai multe features va duce la scăderea erorii de aproximare dar va creste eroarea de estimare și va duce la overfitting.

Folosind mai puține features va reduce eroarea de estimare dar va crește eroarea de aproximare ducând la underfitting.

b). Un model ce conține mai multe ~~labels training sau~~ features va avea o scădere a biasului dar este nevoie de un set de date destul de mare.

Chiar dacă dispun de multe date acestea nu pot oferi informațiile necesare de aceea se folosește feature selection.

Astfel se va reduce overfittingul deoarece nu mai există date redundante.

What is Learning?

What is Learning?

Using Experience
to gain Expertise

“Learning” (in nature): using past experience to make future decisions or guide future actions

Learning in Nature: from Rats to Pigeons

- “learning (in nature)”: the process by which an organism modifies its behavior based on prior experiences.

- different species learn differently, reflecting their innate predispositions or “biases.”

- learning for animals:
  - observing a stimulus (taste, sound, event)
  - associating stimulus with a positive/negative outcome
  - adjusting behavior to avoid danger or gain rewards

Example 1: Rat “Bait Shyness”

Applied Animal Behaviour Science
Volume 24, Issue 2, September 1989, Pages 89-99

ELSEVIER

“Poison-shyness” and “bait-shyness” developed by wild rats (*Rattus rattus* L.). I. Methods for eliminating “shyness” caused by barium carbonate poisoning

Ghazala Naheed, Jamil Ahmad Khan

⊞ **Show more**

https://doi.org/10.1016/0168-1591(89)90037-3

Get rights and content

Abstract

Colonies of wild rats, *Rattus rattus* L., were offered the choice between two baits — cereal grains, flours, mixtures, oily and sweet cereals, and also gram flour. The rats were poisoned in the preferred baits with barium carbonate (100 mg per 10 g food; 200 mg per 10 g food in oily baits) and then presented with the same choice of unpoisoned foods as before.

Poisoning caused a change in the feeding patterns of rats. Foods mixed with barium carbonate were avoided (“poison-shyness”), the same foods then offered without poison were also rejected (“bait-shyness”). Intermittent poisoning also caused aversion to the eating of both poison and bait. Apparently, both the quality and the strength of tastes perceived in the poisonous mixtures influenced the development of “bait-shy” behaviour in the rats.

Example 1: Rat “Bait Shyness”

- rats sample tiny amounts of a new food. If it makes them sick, they avoid it forever.
- one negative experience (nausea) yields a strong taste aversion
- rapid association between taste → toxicity
- rats adapt their behavior, it helps them avoid repeated poisoning
- bait shyness = rats learning to avoid poisonous baits

Example 1: Rat “Bait Shyness”

- learning mechanism for rats: they use past experience with some food to acquire expertise in detecting the safety of the food
- a successful learner should be able to progress from individual examples to broader *generalization* . This is also referred to as ***inductive reasoning*** or ***inductive inference***
- rats apply their attitude on new, unseen examples of food of similar smell and taste

Example 2: Pigeon superstition

Pigeon Superstition: In an experiment performed by the psychologist B. F. Skinner, he placed a bunch of hungry pigeons in a cage. An automatic mechanism had been attached to the cage, delivering food to the pigeons at regular intervals with no reference whatsoever to the birds' behavior. The hungry pigeons went around the cage, and when food was first delivered, it found each pigeon engaged in some activity (pecking, turning the head, etc.). The arrival of food reinforced each bird's specific action, and consequently, each bird tended to spend some more time doing that very same action. That, in turn, increased the chance that the next random food delivery would find each bird engaged in that activity again. What results is a chain of events that reinforces the pigeons' association of the delivery of the food with whatever chance actions they had been performing when it was first delivered. They subsequently continue to perform these same actions diligently.[1]

Example 2: Pigeon superstition

https://www.youtube.com/watch?v=TtfQlkGwE2U

Example 2: Pigeon superstition

"Superstition" in the pigeon.

EXPORT Add To My List © Request Permissions Database: PsycARTICLES Journal Article

Skinner, B. F.

Citation

Skinner, B. F. (1992). "Superstition" in the pigeon. *Journal of Experimental Psychology: General, 121*(3), 273-274.
http://dx.doi.org/10.1037/0096-3445.121.3.273

Abstract

(This reprinted article originally appeared in the *Journal of Experimental Psychology,* 1948, Vol 38, 168–272. The following abstract of the original article appeared in PA, Vol 22:4299.) A pigeon is brought to a stable state of hunger by reducing it to 75% of its weight when well fed. It is put into an experimental cage for a few minutes each day. A food hopper attached to the cage may be swung into place so that the pigeon can eat from it. A solenoid and a timing relay hold the hopper in place for 5 sec at each reinforcement. If a clock is now arranged to present the food hopper at regular intervals with no reference whatsoever to the bird's behavior, operant conditioning usually takes place. The bird tends to learn whatever response it is making when the hopper appears. The response may be extinguished and reconditioned. The experiment might be said to demonstrate a sort of superstition. The bird behaves as if there were a causal relation between its behavior and the presentation of food, although such a relation is lacking.

Example 2: Pigeon superstition

- pigeons receive food at fixed intervals, irrespective of their actions.

- a pigeon that *randomly* spins or pecks at the right moment starts believing that action → reward.

- the pigeon “learns” a superstitious behavior, repeating it, hoping for more food.

- pigeons do not have a strong built-in filter for which correlations are plausible vs. random.

Bait shyness revisited

- **Garcia and Koelling's (1966) Taste Aversion Experiment** explored how rats learn associations between stimuli and negative outcomes.

- rats were given sweetened water to drink. Alongside drinking, two types of additional stimuli were introduced:
  - **bright-noisy stimulus**: a flashing light and a clicking sound.
  - **taste stimulus**: a distinct sweet taste in the water.

- rats were then divided into two groups, each experiencing a different negative consequence:
  - **Group 1 (Nausea)**: After drinking, they were injected with a substance that caused nausea (e.g., lithium chloride).
  - **Group 2 (Electric Shock)**: After drinking, they received a mild electric shock to the feet.

Bait shyness revisited

- **Garcia and Koelling's (1966) Taste Aversion Experiment** explored how rats learn associations between stimuli and negative outcomes.

- **rats who experienced nausea** (Group 1):
  - strongly avoided the **sweet-tasting** water in the future.
  - BUT they did **not** avoid the bright lights or clicking sounds.

- **rats who received shocks** (Group 2):
  - avoided the **bright lights and sounds** after the shock.
  - BUT they continued to drink the **sweetened water** without hesitation.

Bait shyness revisited

- **Garcia and Koelling's (1966) Taste Aversion Experiment** explored how rats learn associations between stimuli and negative outcomes.

- **rats have an innate bias toward associating taste with nausea** (because, in nature, food poisoning comes from ingestion, not from external sounds or lights).

- **rats associate external sensory cues (light/sound) with physical pain** (since, in nature, threats like predators or environmental dangers involve sound or sudden stimuli).

Prior knowledge

- one distinguishing feature between the bait shyness learning and the pigeon superstition is the incorporation of ***prior knowledge*** that biases the learning mechanism = ***inductive bias.***

- the pigeons in the experiment are willing to adopt any explanation for the occurrence of food.

- the rats “know” that water cannot cause an electric shock and that taste is a more reliable indicator of nausea. The rats’ learning process is biased toward detecting some kind of patterns while ignoring other temporal correlations between events.

Prior knowledge

- this experiment illustrates how ***prior knowledge (inductive bias)*** affects learning efficiency.

- in **machine learning**, the equivalent would be defining a **hypothesis space** where only certain correlations are considered meaningful, reducing the risk of "overfitting" to irrelevant patterns.

- just like rats ignore non-relevant associations, ML models need constraints to focus on **real** cause-effect relationships rather than random correlations.

Prior knowledge

- the incorporation of prior knowledge, biasing the learning process, is inevitable for the success of learning algorithms - "No-Free-Lunch theorem"
- the stronger the prior knowledge (or prior assumptions) that one starts the learning process with, the easier it is to learn from further examples.
- the stronger these prior assumptions are, the less flexible the learning is

Mathematical Analysis of Learning

Induction in an urn – a simple learning model

- consider an urn containing a very large number (millions) of marbles, possibly of different types. You are allowed to draw 100 marbles and must predict the distribution of marbles in the urn.

L. G. Valiant, *A theory of the Learnable*, Communications ACM, 27(11):1134-1142, 1984
L. G. Valiant, *Probably Approximately Correct. Nature's Algorithms for Learning and Prospering in a Complex World*, Basic Books, 2013

Induction in an urn – a simple learning model

- consider an urn containing a very large number (millions) of marbles, possibly of different types. You are allowed to draw 100 marbles and and must predict the distribution of marbles in the urn.

- no assumptions
  - impossible task!

- assumption 1: all the marbles are of different types
  - impossible task!

- assumption 2: all the marbles are identical
  - one single draw is sufficient to solve the task!

Induction in an urn (Valiant, 1984)

- assumption 3: 50% of all marbles are of one type
  - probability to miss that type is $(1/2)^{100} = 7.8 * 10^{-31}$
  - predict that after 100 draws you will have seen representatives for 50% of the urn content

- assumption 4: there are at most 5 different marble types
  - **don't know** the distribution of the marbles (***distribution free setting***)
  - could be any distribution: (20%,20%,20%,20%,20%) ,(92%,2%,2%,2%,2%), (49.85%,49.85%,0.1%,0.1%,0.1%), etc
  - we can probably estimate the proportions accurately from 100 samples.

Induction in an urn (Valiant, 1984)

- assumption 3: 50% of all marbles are of one type
  - probability to miss that type is $(1/2)^{100} = 7.8 * 10^{-31}$
  - predict that after 100 draws you will have seen representatives for 50% of the urn content

- assumption 4: there are at most 5 different marble types
  - **don't know** the distribution of the marbles (***distribution free setting***)
  - could be any distribution: (20%,20%,20%,20%,20%) ,(92%,2%,2%,2%,2%), (49.85%,49.85%,0.1%,0.1%,0.1%), etc
  - predict with 97% confidence that after 100 draws you will have seen representatives for more than 80% of the urn content
  - reasoning: (A) if any of the 5 types occurs with frequency > 5%, the probability to miss that type is < $(1\text{-}0.05)^{100}$ = 0.6%. The probability to miss one type is < 5*0.6% = 3%; (not A) There exists types that occurs with frequency < 5%. There can be at most 4 types with frequency < 5% so the rare marble types are < 20%. Probability to miss the common marble types (which account > 80% of the urn) is < 3%.

PAC learning (Valiant, 1984)

- in the urn example, we try to infer the composition of an urn by drawing a small number of marbles. This is analogous to how a learning algorithm tries to generalize from a small training dataset.

- two sources of errors:

  (1) rarity: rare types of marbles are unlikely to be drawn in any small samples

  (2) misfortune: with small probability the sample drawn will be unrepresentative of the contents of the urn because it missed some common marble types

  – neither of these two sources of errors can be totally eliminated BUT we can controlled them by increasing the number of marbles drawn.

- PAC learning: probably approximately correct learning
  – "probably" – misfortune errors
  – "approximately" – rarity errors

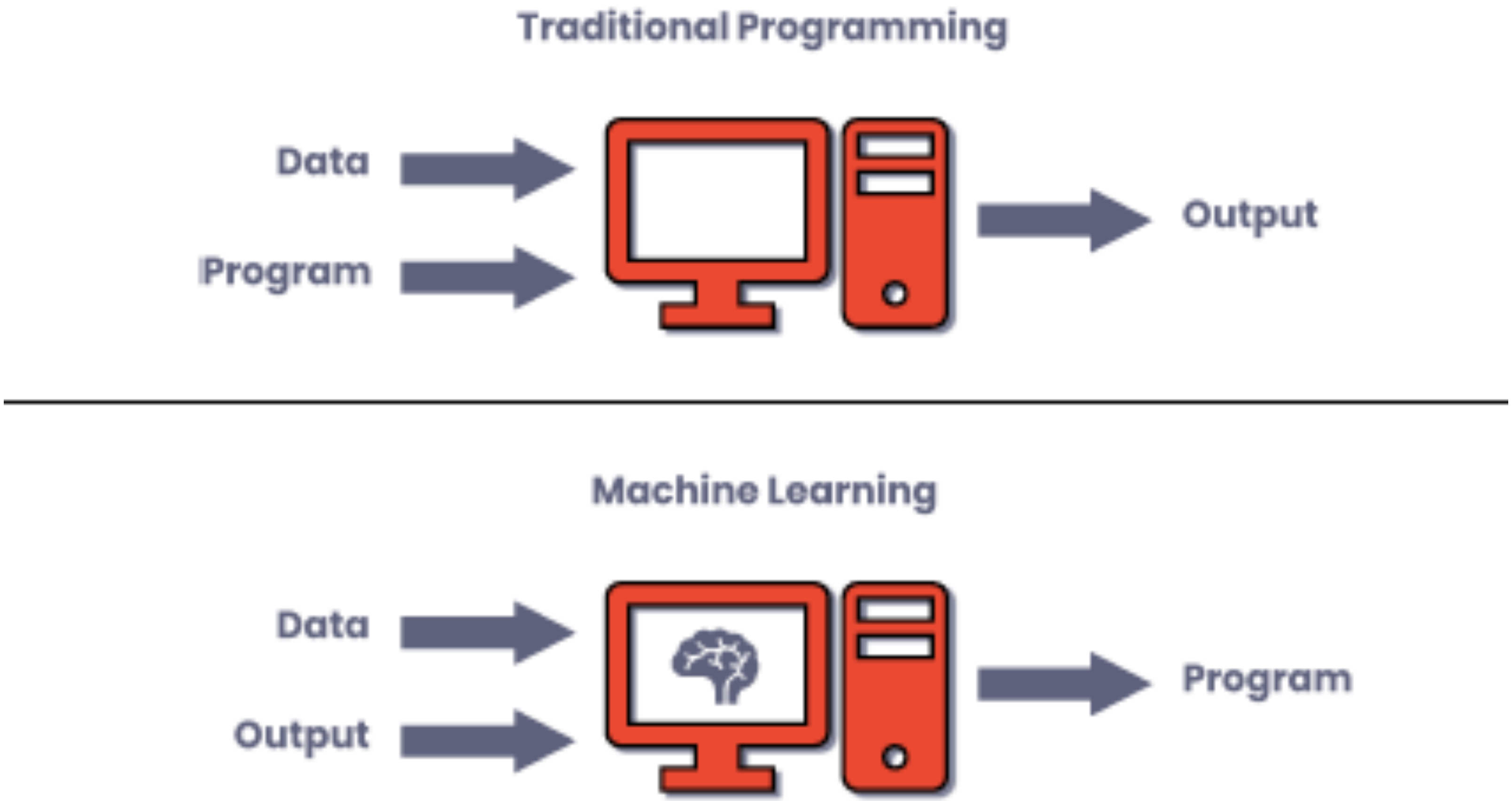
What is Machine Learning?

What is Machine Learning?

Experience (data) $\xrightarrow[\textbf{program}]{\textbf{ML}}$ Expertise (knowledge)

"Machine Learning" as an Engineering Paradigm: Use data and examples, instead of expert knowledge, to automatically create systems that perform complex tasks

What is Machine Learning?



Slide Credit: Sparktech

Machine Learning Everywhere

Spam filtering

Machine translation

Speech recognition

Advertising and ad placement

Recommendation systems

Driving assistance systems

Why do we need machine learning?

- Tasks that are too complex to write a program!

- Computer vision: we know to detect objects but have no idea how we do it!

- Search engines: a human can't read the entire internet!

- Adaptivity and speed of development

Machine Learning vs. Statistics

Machine learning is essentially a form of applied statistics with increased emphasis on the use of computers to statistically estimate complicated functions and a decreased emphasis on proving confidence intervals around these functions

Statistics	Computer Science	Meaning
estimation	learning	using data to estimate an unknown quantity
classification	supervised learning	predicting a discrete Y from $X \in \mathcal{X}$
clustering	unsupervised learning	putting data into groups
data	training sample	$(X_1, Y_1), \ldots, (X_n, Y_n)$
covariates	features	the X_i's
classifier	hypothesis	a map from covariates to outcomes
hypothesis	—	subset of a parameter space Θ
confidence interval	—	interval that contains unknown quantity with a prescribed frequency
directed acyclic graph	Bayes net	multivariate distribution with specified conditional independence relations
Bayesian inference	Bayesian inference	statistical methods for using data to update subjective beliefs
frequentist inference	—	statistical methods for producing point estimates and confidence intervals with guarantees on frequency behavior
large deviation bounds	PAC learning	uniform bounds on probability of errors

Machine Learning vs. Statistics

Differences:

- algorithmic consideration play a major role in ML as we develop algorithms to perform the learning tasks and are concerned with their computational efficiency (*computational complexity)*
- in statistics we are interested in asymptotic behavior (number of training examples grows to infinity) while in ML we work with finite sample sizes (*sample complexity*)
- in ML we consider working under a "distribution-free" setting, where the learner assumes as little as possible about the nature of the data distribution

Course goals

- First goal: to provide a rigorous, yet easy to follow, introduction to the main concepts underlying machine learning:
  - *What is learning?*
  - *How can a machine learn?*
  - *How do we quantify the resources needed to learn a given concept?*
  - *Is learning always possible?*
  - *Can we know if the learning process succeeded or failed?*

- Second goal: present several key machine learning algorithms with strong theoretical foundations

Computational resources of learning

For learning we need 2 type of resources:

1. *Information = training data*
   - analyze in the first part of the course how much training data (sample size) we need in order to learn
   - *sample complexity*

2. *Computation = runtime*
   - for how much time an algorithm (that implements learning) will run, once we have sufficiently many training examples
   - *computational complexity*
   - crucial when we need fast ML applications (driver surveillance, stock exchange trading, etc)
   - runtime = number of elementary instructions executed - arithmetic operations over real numbers - in an asymptotic sense (with respect to input size) of the algorithm, e.g. $O(n)$ *– where n is the size of the input size*

Course Structure – Part 1

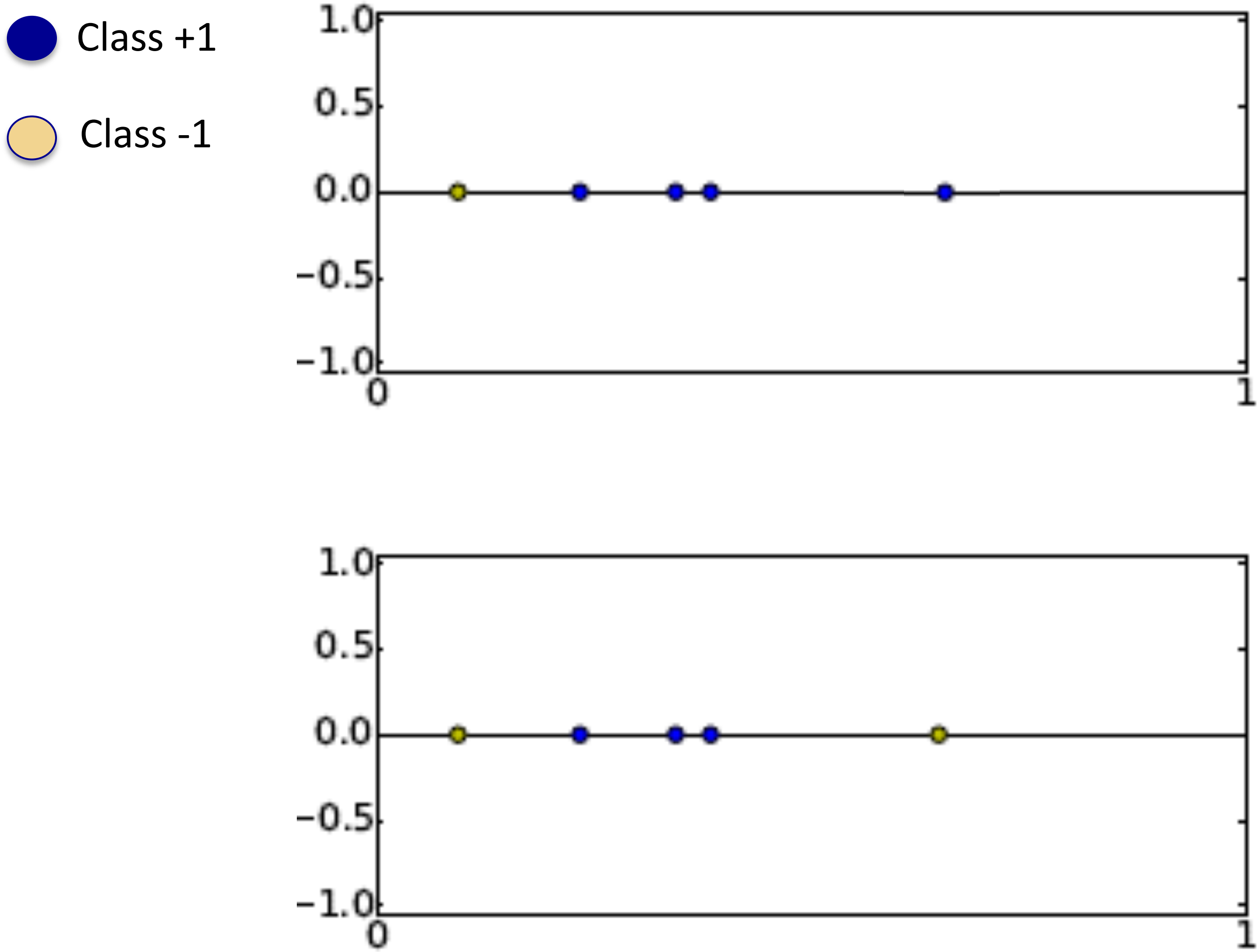
- What is learning?
  - Probably Approximately Correct (PAC) model - Vaillant 1984
- How can a machine learn?
  - Empirical Risk Minimization (ERM)
- Resources needed to learn a given concept?
  - sample complexity, time complexity
- Is learning always possible? Did the learning process succeeded?
  - "no free lunch" theorem

Course Structure – Part 2

- Survey of prominent methods and approaches with strong theoretical foundations such as:
  – Boosting
  – SVMs
  – neural networks? (loose bounds, work in progress)
  – etc

Usefulness of Theoretical Machine Learning

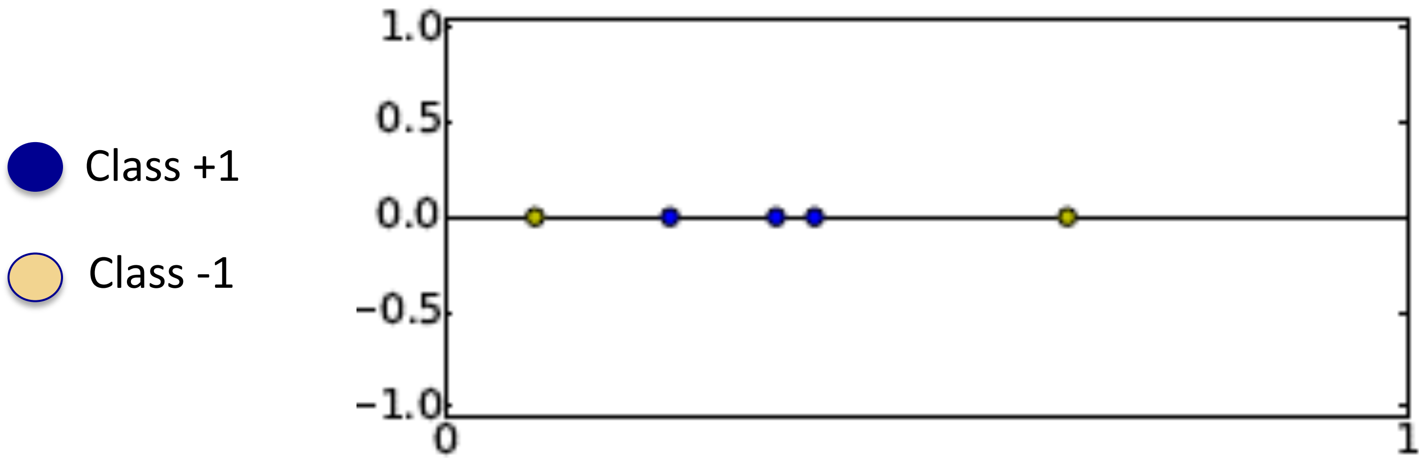
Perceptron in 1D



Linear separable

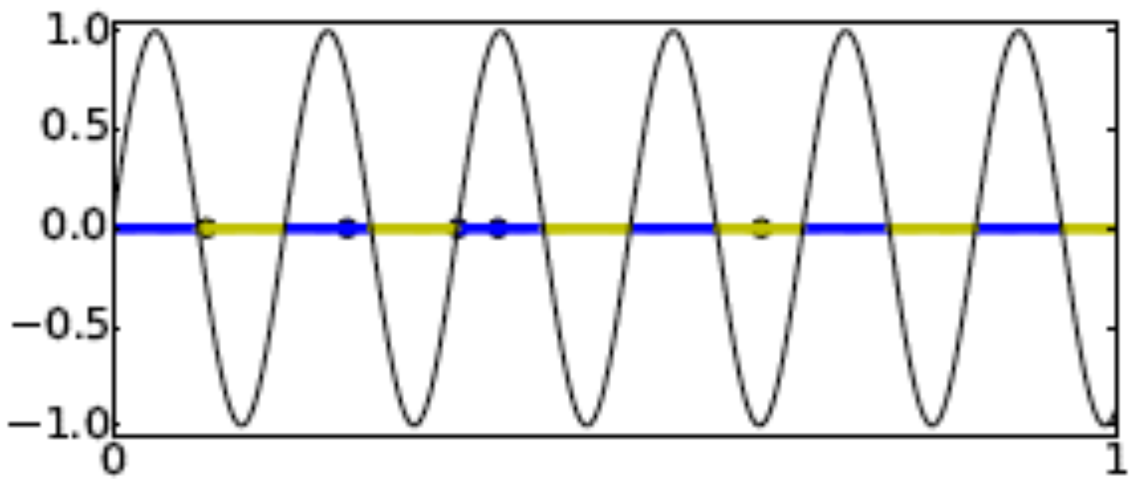
Non-linear separable

Learning sin(λx) in 1D



Non-linear separable

Can you think of an algorithm learning λ for solving the problem?
Will it generalize (small generalization error)?



Theoretical result: cannot learn λ with small generalization error

No Free Lunch theorem

- averaged over all possible data generating distributions, every classification algorithm has the same error rate when classifying previously unobserved points.

- if we make assumptions about the kinds of probability distributions we encounter in real-world applications, then we can design learning algorithms that perform well on these distributions.

- the goal of machine learning research is not to seek a universal learning algorithm or the absolute best learning algorithm. Instead, the goal is to understand what kinds of distributions are relevant to the “real world” that an AI agent experiences and what kinds of machine learning algorithms perform well on data drawn from the kinds of data generating distributions we care about.

Application of Machine Learning: AdaBoost for face detection

ACCEPTED CONFERENCE ON COMPUTER VISION AND PATTERN RECOGNITION 2001

Rapid Object Detection using a Boosted Cascade of Simple Features

Paul Viola
viola@merl.com
Mitsubishi Electric Research Labs
201 Broadway, 8th FL
Cambridge, MA 02139

Michael Jones
mjones@crl.dec.com
Compaq CRL
One Cambridge Center
Cambridge, MA 02142

Abstract

This paper describes a machine learning approach for visual object detection which is capable of processing images

tected at 15 frames per second on a conventional 700 MHz Intel Pentium III. In other face detection systems, auxiliary information, such as image differences in video sequences, or pixel color in color images, have been used to achieve

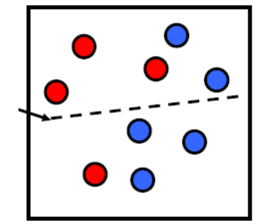
AdaBoost Algorithm

- Given example images $(x_1, y_1), \dots, (x_n, y_n)$ where $y_i = 0, 1$ for negative and positive examples respectively.
- Initialize weights $w_{1,i} = \frac{1}{2m}, \frac{1}{2l}$ for $y_i = 0, 1$ respectively, where m and l are the number of negatives and positives respectively.
- For $t = 1, \dots, T$:
  1. Normalize the weights,
     $$w_{t,i} \leftarrow \frac{w_{t,i}}{\sum_{j=1}^{n} w_{t,j}}$$
     so that w_t is a probability distribution.
  2. For each feature, j, train a classifier h_j which is restricted to using a single feature. The error is evaluated with respect to w_t, $\epsilon_j = \sum_i w_i \, |h_j(x_i) - y_i|$.
  3. Choose the classifier, h_t, with the lowest error ϵ_t.
  4. Update the weights:
     $$w_{t+1,i} = w_{t,i} \beta_t^{1-e_i}$$
     where $e_i = 0$ if example x_i is classified correctly, $e_i = 1$ otherwise, and $\beta_t = \frac{\epsilon_t}{1-\epsilon_t}$.
- The final strong classifier is:

$$h(x) = \begin{cases} 1 & \sum_{t=1}^{T} \alpha_t h_t(x) \geq \frac{1}{2} \sum_{t=1}^{T} \alpha_t \\ 0 & \text{otherwise} \end{cases}$$

where $\alpha_t = \log \frac{1}{\beta_t}$

Start with uniform weights on training examples



$\{\mathbf{x_1, \dots x_n}\}$

For T rounds

Evaluate *weighted* error for each feature, pick best.

Re-weight the examples:
Incorrectly classified -> more weight
Correctly classified -> less weight

Final classifier is combination of the weak ones, weighted according to error they had.

Freund & Schapire 1995

Theoretical Machine Learning

- understand the main concepts underlying machine learning through basic theory
  - the goal is not detailed theory and theorem-proving;
  - emphasis on concepts, less on specific algorithms.
- know the prominent methods used in contemporary machine learning
- learn how to use machine learning *correctly*
- no programming, do ML problems in seminar

Next time

- will cover Chapters 2 and 3 from the book
- do exercises in seminar (from the ones proposed)

Part 1 Foundations 11

2 A Gentle Start 13

2.1 A Formal Model – The Statistical Learning Framework 13
2.2 Empirical Risk Minimization 15
2.3 Empirical Risk Minimization with Inductive Bias 16
2.4 Exercises 20

3 A Formal Learning Model 22

3.1 PAC Learning 22
3.2 A More General Learning Model 23
3.3 Summary 28
3.4 Bibliographic Remarks 28
3.5 Exercises 28